# Empirical Proof Record: VALIDATED

2026-05-24T03:29:05.333826+00:00

## Empirical Proof Record — VALIDATED

**Proof ID:** `8a8009187144992f6bd77d6420e97383…`
**Created:** 2026-05-24T03:29:05.333826+00:00

### 1. Claim

> Across 30 synthetic (x, y) sample pairs of size 50 per group drawn from a rotation of normal, log-normal, and Cauchy distributions with varying shift parameters (seed=20260518), “scipy.stats.mannwhitneyu(use_continuity=True)“ and “pingouin.mwu“ produce Mann-Whitney U statistics AND two-sided p-values that agree to $\leq$ 1e-09 across every pair. (RFC 0002 Phase E1: cross-framework validation; first non-parametric variant.)

- **Operationalisation:** Generate N (x, y) pairs cycling through three distribution shapes (Normal(0,1), LogNormal(0,1), Cauchy(0,1)) with a sweep of location shifts. For each pair, compute Mann-Whitney U + two-sided p under both backends with use_continuity=True. Compute max_abs_diff = max over (pair, statistic $\in$ {U, p}) of |stat_scipy - stat_pingouin|. VALIDATED iff max_abs_diff $\leq$ tolerance.
- **Threshold:** `max_absolute_mannwhitneyu_difference <= 1e-09 U_or_p`
- **H0:** scipy and pingouin disagree on Mann-Whitney U OR the two-sided p-value by more than the floating-point tolerance under matched continuity settings.
- **H1:** Both backends compute identical U AND p to floating-point tolerance across every pair.

### 2. Verdict

**VALIDATED** — observed `0.0`
30 pairs $\times$ 2 backends $\times$ 2 statistics; max pairwise $|\Delta|$ = 0.000e+00 on none ($\leq$ 1e-09 tol)

## 3. Evidence

| Pillar | Statistic | Value | 95% CI | I |
|---|---|---|---|---|
| mann_whitney_u_scipy_vs_pingouin | max_absolute_mannwhitneyu_difference | 0.0 | — | s |

## 4. Pre-registration

- Registered at: 2026-05-24T03:29:05.072800+00:00
- Config hash: ad3e15120cc852c18795179166db8f78…
- Data hash: ad3e15120cc852c18795179166db8f78…

## 5. Signature

af976d6abcc2a90abb07d345b835e9a6140256c4ffb349f7a6cabd48ce3eb164